

ERA5-Augmented Ensemble Learning for Multi-Horizon Tropical Cyclone Track Prediction over the Bay of Bengal: A Leakage-Aware Benchmark

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Abstract

Accurate short- to medium-range cyclone track forecasting is a foundational input to early warning over the Bay of Bengal, a basin that combines complex monsoon-season steering with some of the highest coastal population exposure in the world. We present a rigorously validated benchmark of eight track-prediction systems, a proposed Subset-Expert Context-Aware Ensemble (SECE v2 Phase 3) and seven baselines spanning tree and hybrid families, evaluated at 3, 12, 24, and 48 h lead times on North Indian Ocean IBTrACS storms with genesis in 1940–2024. After quality control the modelling table contains 852 storms and 27,728 three-hourly forecast origins, of which 583 storms and 15,605 origins support multi-horizon supervision. Forty-nine kinematic and derived predictors are augmented with 14 ERA5 reanalysis fields collocated at the storm centre, comprising winds at 850, 500, and 200 hPa, mean sea level pressure, sea surface temperature, a derived deep-layer steering wind, and vertical wind shear. Critically, we identify and correct a subtle form of test-set leakage present in an earlier iteration of our own development, in which architectural decisions were guided by repeated inspection of test-set performance; we replace it with a two-tier protocol in which nine development seeds carry all design decisions and nine disjoint held-out seeds are evaluated exactly once. On the held-out confirm, pooled over 442 distinct test storms, SECE attains median great-circle errors of 4.895, 35.538, 101.186, and 246.331 km. Paired Wilcoxon signed-rank tests with Bonferroni correction across seven comparisons show that SECE is significantly better than every baseline at 3 and 12 h ($p < 0.0071$), statistically indistinguishable from the strongest stacks at 24 and 48 h, and never significantly worse than any baseline at any horizon. Controlled ablations show that ERA5 features deliver genuine but horizon-dependent value, concentrated at 48 h on a per-storm basis, while 5° and 3° spatial patch statistics were tested and not adopted. We report this negative result alongside the positive ones as an honest account of where ensemble sophistication and atmospheric information each contribute.

Keywords: tropical cyclone track prediction; ERA5 reanalysis; ensemble learning; Bay of Bengal; IB-TrACS; evaluation protocol; data leakage.

1 Introduction

The Bay of Bengal has produced a disproportionate share of the deadliest tropical cyclones in recorded history. Its shallow bathymetry, funnel geometry, and low lying deltaic margin convert moderate storms into severe surge events, and the population density along that margin means that forecast track error translates directly into evacuation scope and timing [3, 4, 16]. Reliable guidance at lead times of a few hours to two days is therefore not an academic refinement but an operational requirement.

Numerical weather prediction remains the reference standard for that guidance, and dynamical-model consensus is a mature part of operational practice [19, 12]. Its computational and assimilation demands, however, are frequently beyond the reach of national meteorological agencies in devel-

oping countries. Data-driven approaches that learn directly from historical best-track archives offer a lightweight complement: the archives are public and well-curated [1, 2], and global reanalysis now supplies a physically consistent reconstruction of atmospheric state at every historical storm position [27], so a learned model can condition on environmental context without running a forecast cycle [10, 22].

Published approaches span a wide methodological range. Climatological-persistence analogues of the CLIPER family remain standard references [15, 18]; recurrent and convolutional sequence models have been applied extensively to track prediction [8, 9, 21]; and transformer architectures show promise at larger scales [23]. At the sample sizes available for a single basin, where storm counts are in the hundreds rather than the millions, tree ensembles over

engineered features nonetheless remain difficult to displace [14, 13].

Three substantive gaps persist specifically for the North Indian Ocean. First, architectural benchmarking is narrow: most studies compare a small number of model families, so claims of relative superiority rest on thin comparisons [11, 20]. Second, evaluation is usually confined to a single lead time, which leaves horizon-dependent behaviour undocumented even though the operational value of a forecast changes sharply with lead [22]. Third, the incremental contribution of atmospheric reanalysis beyond kinematic features has not, to our knowledge, been isolated for this basin by controlled single-variable ablation.

A fourth gap is methodological, and it motivates a central contribution of this paper. Published ensemble benchmarks rarely disclose, or guard against, the leakage introduced by iterative human-guided architecture search. When a designer repeatedly inspects held-out performance and adds components that close the gaps observed there, the architecture has been fitted to the test set even though no model parameter ever saw a test label, and the resulting held-out estimate is optimistic [36]. We encountered exactly this failure mode in our own development, and we document it here together with the diagnostic that exposed it and the protocol that replaced it.

1.1 Contributions

- An eight-system benchmark evaluated at four forecast horizons over the Bay of Bengal and wider North Indian Ocean, under storm-wise splits, with bootstrap confidence intervals and paired significance testing pooled over nine held-out seeds and 442 distinct test storms.
- SECE v2 Phase 3, an ensemble that combines physics-informed feature subsets, heterogeneous tree learners, and a dedicated ERA5-derived atmospheric expert, fused through horizon-specific non-negative least squares candidates selected by a validation-only router.
- A controlled, single-variable-at-a-time ablation isolating the effect of point ERA5 features against track-only kinematics, of a dedicated atmospheric expert against dispersing ERA5 into existing subsets, and of spatial patch features against single-point extraction.
- Full statistical validation rather than point-estimate comparison, together with a transparent account of the leakage failure mode encountered during development and the corrected protocol adopted in response.

2 Related Work

Classical statistical methods form the baseline layer of operational cyclone forecasting. Climatology-and-persistence models of the CLIPER family remain reference systems against which skill is measured [15, 18], and statistical-dynamical methods have long been applied to Bay of Bengal storms alongside surge modelling [16]. Dynamical-model consensus, in which several independent forecasts are averaged, is the operational expression of the observation that combining diverse predictors reduces track error [19, 12]; the ensembles studied here can be read as a learned analogue of that idea.

Machine learning for cyclone trajectories has largely followed the deep learning trajectory of neighbouring fields. Recurrent architectures have been applied to hurricane and typhoon tracks [9, 21], convolutional-recurrent hybrids with multi-dimensional feature selection have been proposed for typhoon tracks [8], fused deep learning over aligned reanalysis fields has been demonstrated for track forecasting [10], and non-iterative spatiotemporal transformers have recently been applied to intensity [23]. Reviews of ensemble deep learning [13] and of machine learning for cyclone track and intensity [22] survey this space.

Against that trend, Grinsztajn et al. [14] show that tree-based methods consistently outperform deep learning on medium-sized tabular data with engineered features, a regime that describes basin-scale cyclone track prediction closely: storm counts here are in the hundreds, and the predictive signal is concentrated in a few dozen hand-constructed kinematic and environmental quantities. Our results extend that observation one level further. Even within the tree-based ensemble family, additional architectural complexity does not improve performance uniformly as lead time grows; a simpler stacking approach becomes statistically competitive with a more elaborate context-aware ensemble at longer leads.

Basin-specific work on the North Indian Ocean has tended to address either intensity classification from best-track data [11], single-storm case studies with dynamical models [20], or neural regression for track at a single horizon [28]. Geographically focused machine-learning studies elsewhere reinforce the value of basin-specific modelling [29], and recent global benchmarking efforts underline how much evaluation protocol matters when systems are compared [7]. To our knowledge no prior published benchmark evaluates this breadth of model families across four horizons on Bay of Bengal data under a leakage-controlled, statistically validated protocol.

3 Data and Experimental Design

3.1 Source archives and study domain

Storm trajectories come from the International Best Track Archive for Climate Stewardship, version 4, North Indian Ocean basin [1, 2]. Atmospheric state comes from ERA5, the fifth-generation ECMWF re-analysis, which provides physically consistent gridded fields at sub-daily resolution continuously back to 1940 [27] and is the source of every non-track-derived feature used in this study.

ERA5 fields were extracted over a domain of 30°N–5°N, 70°E–102°E at three-hourly resolution for all calendar days on which a tracked storm was active, spanning 551 storm-months and approximately 10.5 GB of source NetCDF, and joined to the quality-controlled track rows.

3.2 Quality control and sample counts

We retain North Indian Ocean storms with genesis dated 1940–2024, applying quality-control criteria that remove records with irregular time steps, coordinate inconsistencies, and track merge or spur artefacts. This yields 852 storms and 27,728 three-hourly track points, the large majority originating in the Bay of Bengal proper with the remainder from Andaman and Arabian Sea genesis.

Multi-horizon target construction, which requires a verified position 12, 24, and 48 h ahead of each origin, imposes a minimum sustained track length. This length requirement, and not a second quality-control pass, excludes 269 short-lived storms and leaves a modelling pool of 583 storms and 15,605 multi-horizon samples. Because train, validation, and test assignment is determined once at the multi-horizon level, all four horizons including 3 h draw from this same storm pool, which keeps the comparison across horizons internally consistent.

3.3 Storm-wise partitioning

All partitioning is by unique storm identifier rather than by individual observation. This matters because successive positions within one trajectory are strongly autocorrelated, so an observation-level split would place near-duplicate rows on both sides of the partition boundary and inflate apparent skill. For each seed, storms are drawn 70:15:15 into train, validation, and test, giving 408, 87, and 88 storms respectively. Sample counts vary across seeds with the length distribution of the storms drawn, as Table 1 shows.

For significance testing we pool held-out test predictions across all nine held-out seeds. The figure

of 442 quoted throughout is the number of *distinct* storm identifiers appearing in that pooled test sample; because a storm may fall in the test partition under more than one seed, this is smaller than the sum of the per-seed test counts. Pooled median errors use all pooled test points, while paired per-storm medians are defined on these 442 storms.

3.4 Feature engineering

Track and physics features (49), used by all variants. Storm position and lagged positions up to three steps back; motion kinematics comprising displacement, translation speed, bearing with sinusoidal encoding, acceleration, and velocity trend; derived physical proxies comprising bearing curvature and stability, seasonal encoding, and a latitude-and-bearing recurvature proxy; native environmental fields comprising distance to land, landfall state, and wind estimate; and indicator flags marking imputed values. This construction follows established practice for kinematic track modelling [28, 8].

ERA5 point features (14), added in the augmented configuration. Zonal and meridional wind at 850, 500, and 200 hPa; mean sea level pressure; sea surface temperature; a derived deep-layer steering wind in both components; and derived vertical wind shear as the 200 hPa minus 850 hPa difference in both components together with its magnitude.

Sea surface temperature requires specific comment. ERA5 leaves it undefined over land, so near-coast track points have no valid value. We fill these from the nearest valid ocean cell and retain a binary flag marking the substitution. Approximately 38% of rows required this correction, and the affected rows concentrate at higher latitude, more easterly, near-coast positions, which is the signature of a land-masking artefact rather than a bounding-box or data-collection defect. Retaining the flag as a model input lets the learner condition on the substitution rather than treating imputed and observed values identically.

A supplementary spatial-patch feature set was also extracted, adding six features per patch size that capture wind-field shear gradient, asymmetry, and nearest-trough proximity within 5° and 3° boxes centred on the storm. It was evaluated but not adopted, for the reasons given in Section 6.1.

Because the subset-expert design assumes these engineered features carve the data into physically meaningful regimes rather than arbitrary numerical groupings, we verify that assumption before modelling. Fig. 1 projects the 63-dimensional space onto two dimensions with t-SNE [31] on a 10,000-sample subset, coloured independently by storm nature, translation speed, latitude band, and displacement magnitude. Coherent structure under all four colourings, rather than under only one, indicates

Table 1: Held-out split composition. Storm counts are fixed by the 70:15:15 storm-wise draw; sample counts vary with the track-length distribution of the storms assigned. “MH” denotes multi-horizon samples supporting 12, 24, and 48 h targets.

Seed	Storms			MH samples			3 h samples		
	Train	Val	Test	Train	Val	Test	Train	Val	Test
0	408	87	88	11,066	2,169	2,370	17,594	3,561	3,778
1	408	87	88	11,067	2,425	2,113	17,595	3,817	3,521
2	408	87	88	11,163	2,223	2,219	17,691	3,615	3,627
7	408	87	88	10,352	2,679	2,574	16,880	4,071	3,982
13	408	87	88	11,120	2,262	2,223	17,648	3,654	3,631
99	408	87	88	11,269	1,930	2,406	17,797	3,322	3,814
123	408	87	88	10,693	2,387	2,525	17,221	3,779	3,933
2024	408	87	88	10,730	2,313	2,562	17,258	3,705	3,970
2026	408	87	88	11,081	2,074	2,450	17,609	3,466	3,858

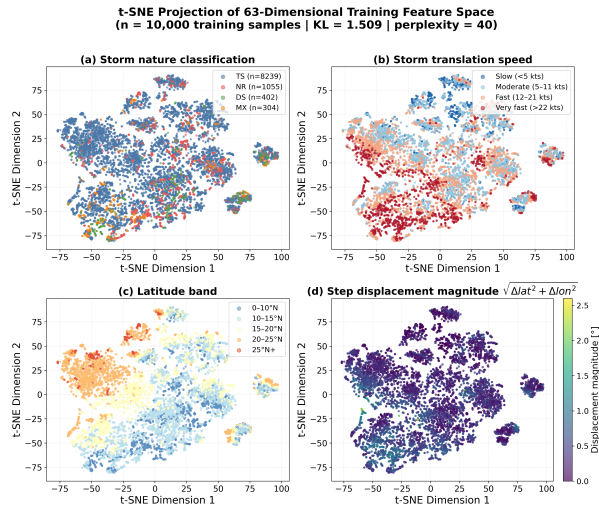


Figure 1: t-SNE projection [31] of the 63-dimensional training feature space on a 10,000-sample subset, coloured by independent meteorological attributes. Coherent separation across colourings indicates that the engineered features encode physically distinct regimes, which is the premise underlying the subset-expert design.

that the feature space separates distinct kinematic and environmental regimes, which is the premise the subset experts of Section 5 exploit.

3.5 Genesis-window selection

The choice of genesis window trades training-set size against observational quality. We evaluated three candidates, 1940–2024, 1979–2024, and 1990–2024, on nine development seeds using a simple LightGBM baseline. Because both IBTrACS and ERA5 quality improve substantially once satellite observation begins, and because pre-1979 records show near-total missingness in the wind observations used for intensity estimation, a naive comparison risks confounding “more training data” with “easier test set”. We therefore added a fixed-test control in which models trained on each candidate window are

all evaluated on one common modern holdout drawn from 1990–2024 genesis, with the same test storms excluded from every training set.

Table 2 gives both comparisons. The wide window wins at every horizon under the fixed-test control, so its advantage is not merely a test-composition artefact. The size of that advantage is nonetheless strongly horizon-dependent: at 3 h the raw gap of 0.90 km falls to 0.18 km once test composition is held fixed, whereas at 24 and 48 h gaps of 7.9 and 17.7 km persist. We attribute this to storm *position* records remaining comparatively reliable across eras even where wind and intensity observations are sparse, which is the relevant consideration because track geometry, not intensity, is the target variable here. We accordingly lock 1940–2024 for all subsequent modelling.

4 On Test-Set Leakage: A Methodological Note

During initial development, successive architecture revisions, which we refer to internally as Phases 1 through 3, were evaluated on a single fixed train/validation/test split. The decision to advance from one phase to the next was taken by inspecting per-horizon performance gaps on that test set and introducing targeted fusion-path additions intended to close them. No individual test observation was used to fit any model parameter. Nevertheless this iterative, human-guided design loop constitutes test-set leakage: the architecture itself was shaped by repeated exposure to test-set feedback, which invalidates the resulting held-out estimate as a measure of generalisation [36].

We detected the pattern by auditing, for each architectural decision point, which data partition had informed it. Within-run decisions, namely fusion-path selection and non-negative least squares weight fitting, proved to be properly restricted to the validation partition. The *cross-run* decision to advance from one phase to the next, however, had been made

Table 2: Genesis-window comparison, mean test median error in km over nine development seeds with a simple LightGBM baseline. The upper block lets each window use its own test split; the lower block trains on each window and evaluates all three models on one common 1990–2024 genesis holdout.

Training window	3 h	12 h	24 h	48 h
<i>Each window on its own test split</i>				
1940–2024	5.24	36.19	99.77	244.92
1979–2024	6.13	44.04	112.38	271.40
1990–2024	6.14	44.11	108.12	261.58
<i>Fixed common modern holdout</i>				
1940–2024	5.96	39.42	100.18	243.86
1979–2024	6.07	42.24	103.86	249.82
1990–2024	6.14	44.11	108.12	261.58

by inspecting test-set rank tables. The empirical symptom was unambiguous: an architecture that appeared to win at every horizon on the original split did not reproduce that sweep on independent fresh splits.

Correction. We adopted a strict two-tier seed protocol. Nine *development* seeds, {3, 4, 5, 6, 8, 9, 10, 11, 14}, carry every architectural and fusion-path decision; during this stage the corresponding held-out partition is neither generated nor examined. Nine *held-out* seeds, {0, 1, 2, 7, 13, 99, 123, 2024, 2026}, were generated once and evaluated exactly once, after the architecture was frozen. Architectural convergence is verified by confirming that development-seed and held-out-seed results agree, reported in Section 6.2, rather than by any further tuning against held-out performance.

We report this process transparently for two reasons. Iterative test-set inspection is, in our reading, a common and under-scrutinised risk in ensemble-based forecasting benchmarks. And our own uncorrected results, which appeared to show the proposed ensemble winning at every horizon, failed to replicate, which makes the risk concrete rather than hypothetical.

5 Methodology

5.1 Baseline systems

Seven baselines are benchmarked alongside the proposed ensemble. The **Stacking Ensemble** combines Random Forest, XGBoost, and LightGBM base learners through a Ridge-penalised, cross-validated linear meta-learner. The **Persistence Residual Cascade (PRC)** corrects a naive persistence extrapolation with a dedicated LightGBM residual model fitted in kilometre space and mapped back to degrees. **Random Forest** [30], **LightGBM** [25], **XGBoost** [24], and **CatBoost** [26] are included as individual learners. **CB+MotionNN** corrects a CatBoost predictor with a 64-32-16 feed-forward network operating on kinematic residual

features.

5.2 Proposed architecture: SECE Phase 3

For each forecast horizon, SECE constructs four physics-informed feature subsets. At 3 h these are position, motion, environ, and the full feature set; at 12, 24, and 48 h they are motion, physics, environ, and full. Each subset is modelled independently by four tree algorithms, namely CatBoost, XGBoost, LightGBM, and Random Forest, giving sixteen subset-level models per horizon, which are fused by non-negative least squares into a single subset-consensus signal with separate weights for the latitude and longitude components.

That subset-consensus signal is then combined with the seven baseline predictions, the “champion” pool, in a second-stage non-negative least squares fusion. The router never sees held-out test data: for each seed and horizon it evaluates a *fixed menu* of fusion paths on the validation storms only and locks the path with the lowest validation median error. At 3 h the menu includes full SECE NNLS over all champions, NNLS over the champion subset alone, Random-Forest-plus-stacking NNLS, and stacking with a LightGBM residual correction in kilometre space. At 12 h only the two NNLS variants over the full pool and over champions are eligible, reflecting the shorter list of gains observed once ERA5 environ features were locked. At 24 and 48 h the menu adds persistence-anchored kilometre-space NNLS over champions—weights are fit on displacement residuals measured in kilometres and the persistence term is added back after fusion—and at 48 h further adds LightGBM+PRC NNLS and its kilometre-space analogue, because long-lead development ablations showed residual-cascade corrections to be competitive with pure stacking. The selected path’s predictions on the disjoint test storms are what we report in Section 6.2. Both router selection and every NNLS weight vector are restricted to the validation partition, in keeping with Section 4. Fig. 2 summarises the pipeline.

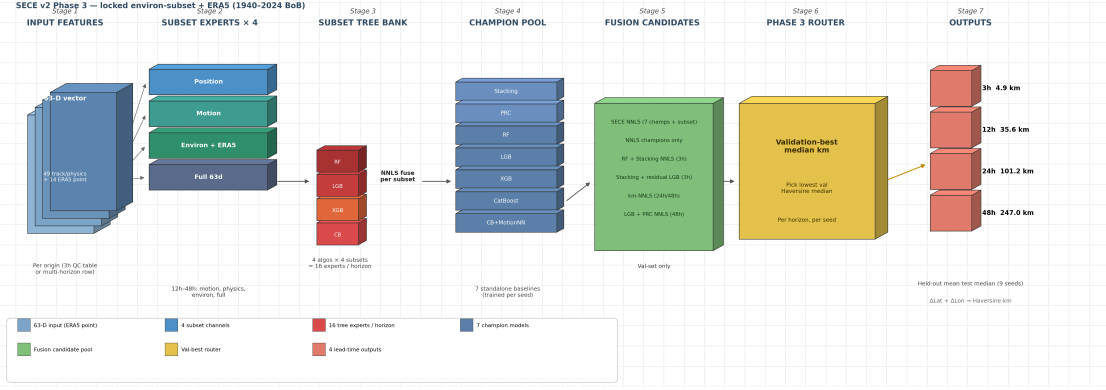


Figure 2: Locked SECE v2 Phase 3 pipeline. Sixty-three inputs, 49 kinematic and derived features plus 14 ERA5 point features, feed four horizon-specific subset experts, each modelled by four tree algorithms and fused by non-negative least squares into a subset-consensus signal. That signal joins the seven champion baselines in a second-stage fusion pool, and a validation-only router selects the final candidate per seed.

The environ expert. Rather than folding the 14 ERA5 features into existing subsets, we allocate them, together with the native environmental fields for distance to land, landfall state, and storm speed and direction, to a dedicated environ subset with its own four-algorithm expert pool. This choice was made over three tested alternatives, namely ERA5 features added directly to existing point-feature subsets and ERA5 spatial-patch features at 5° and 3° , on the basis of the best balanced performance across all four horizons reported in Section 6.1. It also admits a more direct mechanistic reading, since atmospheric information receives explicit ensemble representation rather than competing for splits inside a larger undifferentiated subset.

5.3 Training protocol

All tree models use the settings in Table 3, applied uniformly across every system so that measured differences reflect architecture rather than tuning budget [14]. Early stopping is disabled for the tree learners.

5.4 Evaluation metric and statistical testing

The primary metric is median great-circle (Haversine) error in kilometres between predicted and verified storm position, computed per test storm and aggregated [40]. The median rather than the mean is used because per-storm error distributions are strongly right-skewed, so a mean would be dominated by a small number of recurving or rapidly accelerating systems.

For the final held-out evaluation we additionally report 95% confidence intervals on the pooled median obtained from 10,000 storm-level bootstrap resamples [37], and paired Wilcoxon signed-rank tests comparing SECE against each of the seven baselines

on matched per-storm median errors [38]. Resampling and pairing are performed at storm level, not sample level, so that the unit of statistical independence matches the unit of the split. Bonferroni correction is applied across the seven simultaneous comparisons [39], giving an adjusted threshold of $\alpha = 0.05/7 \approx 0.0071$.

5.5 Systems excluded from the held-out table

Before the ERA5-augmented design was frozen we trained CNN-GRU and bidirectional LSTM [8, 34] alongside the tree learners on storm-wise development splits, using the same fixed budget philosophy as the locked tree protocol (300 estimators or epochs, learning rate 0.01, no architecture-specific hyperparameter search). In those exploratory runs—track-only features first, then informal checks after ERA5 was added—both recurrent models repeatedly ranked below Random Forest, stacking, and gradient boosting at 3, 12, and 24 h, with short-lead errors closer to persistence-scale displacement than the sub-6 km medians the trees achieved. That pattern is consistent with Grinsztajn et al. [14] on tabular cyclone features at basin sample sizes. We therefore scoped the leakage-corrected held-out confirm to the eight tree and hybrid systems already in the production pipeline (Table 5) rather than porting sequence training to nine fresh held-out seeds and a 48 h head. CNN-GRU and BLSTM are absent from every held-out result and figure reported here.

6 Results

6.1 Development-phase ablation

Table 4 reports, for each feature configuration, the number of development seeds out of nine on which

Table 3: Locked hyperparameters, applied uniformly across systems unless noted.

Component	Setting
Tree estimators (RF, XGB, LGB, CatBoost)	300
Maximum depth (CatBoost: depth)	6
Global learning rate	0.01
LightGBM <code>num_leaves</code>	63
XGBoost <code>subsample</code> / <code>colsample by tree</code>	0.8 / 0.8
Stacking Ridge meta-learner	150 cross-validated meta-samples
MotionNN	MLP 64-32-16, ReLU, lr 0.01, max 100 iterations, early stopping on
PRC residual model	LightGBM on persistence residual, km space

SECE achieved the lowest median error among all eight systems, together with its mean median error across those seeds.

Two findings motivate the locked architecture. First, ERA5 point features substantially reduce SECE’s mean error at 48 h, from 250.26 to 246.80 km, with 8 of 9 development seeds improving. This establishes that atmospheric information carries genuine, exploitable signal at long lead times rather than merely adding dimensionality. Second, allocating those features to a dedicated environ expert instead of embedding them in existing subsets improves 24 h performance specifically, raising rank-1 frequency from 2/9 to 3/9 and reducing mean error from 99.86 to 99.43 km; 24 h was precisely the horizon at which the plain point-ERA5 configuration was weakest.

The same table records a cost that we report rather than suppress. At 3 h the track-only configuration retains the lowest mean error, 4.84 against 4.93 km, so ERA5 does not help uniformly across horizons. The short-range gain from ERA5 in this architecture is expressed as rank stability rather than as a lower mean.

Spatial-patch features improved short-range rank-1 frequency, reaching 7/9 at both 3 and 12 h, but showed no consistent advantage over the environ-expert configuration at 24 or 48 h, where means fall within roughly 0.4 km of each other, inside the seed-to-seed noise floor. We therefore did not adopt spatial-patch features into the locked architecture, and report this as a negative result.

6.2 Held-out evaluation

Table 5 reports pooled median errors and bootstrap confidence intervals for all eight systems at all four horizons, over 34,114 test samples at 3 h and 21,442 at the longer horizons.

At 3 h SECE leads with 4.895 km, ahead of Random Forest at 4.927 km and the Stacking Ensemble at 4.945 km, with a clear gap to the remaining five systems. At 12 h SECE again leads with 35.538 km against 36.146 km for stacking. At 24 h the ordering changes: stacking and XGBoost take the top two point estimates, 100.771 and 101.029 km,

with SECE third at 101.186 km. At 48 h XGBoost and PRC lead with 242.702 and 243.229 km against 246.331 km for SECE. In both long-lead cases the confidence intervals overlap almost entirely, which is why the paired tests rather than the point estimates carry the inferential weight.

Table 6 gives those tests in full, with Bonferroni-adjusted p -values and the rank-biserial correlation as an effect size. At 3 and 12 h all seven comparisons are significant after correction, so SECE beats every baseline at both short horizons. At 24 h only CatBoost and CB+MotionNN remain significant; Random Forest is significant raw but not after correction, and stacking, PRC, LightGBM, and XGBoost are not significant by either standard. At 48 h only CatBoost remains significant, with stacking and CB+MotionNN significant raw but not after correction.

The effect-size column carries an important detail. Every rank-biserial correlation in the table is negative, which under our sign convention means SECE holds the lower per-storm error in the paired comparison. Consequently SECE is never significantly *worse* than any baseline at any horizon; at long lead the correct statement is parity, not defeat. The magnitudes also make the horizon dependence visible: against stacking the effect size falls from -0.208 at 3 h to -0.071 at 24 h, and against XGBoost from -0.993 to -0.003 at 48 h.

Pooled medians summarise central tendency but hide seed-to-seed variability. Across the nine held-out splits, SECE achieves the lowest test median on six seeds at both 3 and 12 h; Random Forest leads on two seeds at 3 h and stacking on one. At 12 h stacking wins on the remaining three seeds. At 24 h no system dominates: stacking, LightGBM, and XGBoost each lead on two seeds and SECE on two, with the pooled ordering therefore sensitive to which storms land in test under each draw. At 48 h PRC leads on four seeds, XGBoost on three, SECE on one, and stacking on one—consistent with the pooled table where PRC and XGBoost sit slightly ahead of SECE even though Wilcoxon tests do not declare a significant SECE *deficit*.

Read together, Table 5, the seed counts above,

Table 4: Development-seed ablation over nine development seeds. Each cell gives the SECE rank-1 count out of nine, then the mean median error in km. The locked configuration is in bold. The final row is the independent held-out confirm on disjoint seeds, included here only to show tier agreement.

Configuration	3 h	12 h	24 h	48 h
Track-only (no ERA5)	5/9, 4.84	3/9, 35.48	3/9, 100.48	0/9, 250.26
+ ERA5 point features	5/9, 4.93	7/9, 35.59	2/9, 99.86	1/9, 246.80
+ environ expert (locked)	5/9, 4.93	6/9, 35.56	3/9, 99.43	2/9, 246.94
+ spatial patch (5°)	7/9, 4.93	7/9, 35.56	1/9, 99.74	1/9, 246.55
+ spatial patch (3°)	7/9, 4.93	7/9, 35.66	2/9, 99.42	3/9, 247.00
Held-out confirm (locked)	6/9, 4.90	6/9, 35.63	2/9, 101.21	1/9, 246.99

Table 5: Held-out test performance ranked at each horizon: model and median great-circle error (km) with 95% bootstrap confidence interval from 10,000 storm-level resamples. Nine held-out seeds pooled (442 distinct storm identifiers; 34,114 test samples at 3 h and 21,442 at 12–48 h). Eight tree/hybrid systems only; CNN-GRU and BLSTM omitted (Section 5.5).

Rank	3 h Horizon			12 h Horizon			24 h Horizon			48 h Horizon		
	Model	Med.	95% CI	Model	Med.	95% CI	Model	Med.	95% CI	Model	Med.	95% CI
1	SECE	4.895	4.71–5.08	SECE	35.538	34.2–36.9	Stacking	100.771	96.9–104.7	XGBoost	242.702	232.6–255.4
2	RF	4.927	4.73–5.13	Stacking	36.146	34.7–37.7	XGBoost	101.029	96.9–105.3	PRC	243.229	231.7–255.2
3	Stacking	4.945	4.76–5.12	RF	36.901	35.5–38.3	SECE	101.186	97.3–104.8	LGB	245.975	234.9–257.2
4	CB+MoNN	5.813	5.70–5.93	LGB	36.928	35.5–38.3	LGB	101.632	97.8–105.6	Stacking	245.977	235.9–257.2
5	LGB	6.026	5.89–6.17	XGB	36.947	35.5–38.4	PRC	101.941	98.1–105.8	SECE	246.331	235.5–257.6
6	XGB	6.411	6.24–6.58	PRC	37.177	35.7–38.5	RF	103.511	99.7–107.4	CB+MoNN	249.678	239.0–261.2
7	CatBoost	8.479	8.28–8.69	CB+MoNN	37.413	36.0–38.9	CB+MoNN	104.034	100.2–108.1	RF	251.305	240.8–262.3
8	PRC	8.872	8.71–9.06	CatBoost	42.028	40.4–43.7	CatBoost	107.191	102.6–111.8	CatBoost	254.862	243.0–267.8

Table 6: Paired Wilcoxon signed-rank tests of SECE against each baseline on 442 matched per-storm medians, reporting the Bonferroni-adjusted p -value and the rank-biserial correlation r as an effect size. Bold marks a significant SECE advantage at the adjusted threshold $\alpha \approx 0.0071$; a dagger marks significance before correction only. Every r is negative, meaning SECE holds the lower per-storm error, so no baseline is significantly better than SECE at any horizon. The collapse of $|r|$ from 3 to 48 h is the quantitative signature of the horizon dependence discussed in Section 7.

Baseline	3 h		12 h		24 h		48 h	
	p	r	p	r	p	r	p	r
Stacking Ensemble	0.0016	−0.208	0.0014	−0.210	1.000	−0.071	0.262 [†]	−0.114
Random Forest	0.0272	−0.164	1.1×10^{-5}	−0.263	0.0737 [†]	−0.140	0.704	−0.090
LightGBM	$< 10^{-6}$	−0.970	$< 10^{-6}$	−0.306	1.000	−0.044	1.000	−0.068
XGBoost	$< 10^{-6}$	−0.993	$< 10^{-6}$	−0.364	1.000	−0.046	1.000	−0.003
PRC	$< 10^{-6}$	−1.000	$< 10^{-6}$	−0.394	0.898	−0.084	1.000	−0.005
CatBoost	$< 10^{-6}$	−1.000	$< 10^{-6}$	−0.806	$< 10^{-6}$	−0.450	$< 10^{-6}$	−0.352
CB+MotionNN	$< 10^{-6}$	−0.833	$< 10^{-6}$	−0.443	1×10^{-5}	−0.265	0.262 [†]	−0.114

and Table 6 imply a horizon-dependent picture. After pooling all nine held-out seeds (442 distinct storms), SECE is the lowest-median system at 3 and 12 h (4.895 and 35.538 km), leads on six of nine seeds at both horizons, and is significantly better than every baseline under Bonferroni correction ($\alpha \approx 0.0071$). At 24 h stacking is the pooled leader (100.771 km versus SECE 101.186 km, rank 3); SECE leads only two seeds, and paired tests show parity with stacking rather than a significant SECE deficit. At 48 h XGBoost leads the pool (242.702 km versus SECE 246.331 km, rank 5),

SECE wins on one seed only, and Wilcoxon again indicates parity with XGBoost. Short-lead advantage is therefore both pooled and seed-stable; long-lead guidance is better described as consensus among strong tree stacks than as a single dominant learner.

Fig. 3 places the eight systems in the 3 h error–accuracy plane (mean on the horizontal axis; medians in Table 5), where the three leading systems separate visibly from the remainder.

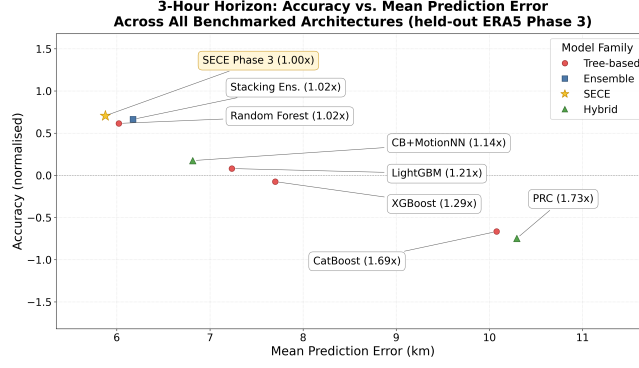


Figure 3: Held-out 3 h benchmark for the eight locked systems (pooled nine held-out seeds). Horizontal axis: mean great-circle error (km); vertical axis: normalised accuracy from median error relative to the tree-based family median. Pooled medians and bootstrap intervals are in Table 5.

6.3 Per-storm ERA5 contribution at long horizons

To characterise the distribution of ERA5’s benefit independently of the ensemble machinery, we compare a single LightGBM model trained with and without ERA5 features, storm by storm, on a representative development seed with 88 test storms. At 24 h ERA5 improves accuracy for 43 of 88 storms, or 48.9%, with a near-zero mean effect, indicating rough parity at this horizon for this simpler model. At 48 h it improves accuracy for 54 of 88 storms, or 61.4%, with a median per-storm improvement of 10.9 km.

The benefit therefore strengthens with lead time. This is consistent with the mechanistic expectation that a storm’s own recent kinematic history becomes a progressively weaker predictor of its future position, so large-scale steering information becomes proportionally more valuable [10, 12].

6.4 Feature importance within the environ expert

Table 7 reports gain-based importance for the ERA5 fields inside the environ-subset expert, from a diagnostic retrain on one held-out seed under the locked LightGBM specification, averaged across the latitude and longitude output heads. This is a single-seed tree-gain diagnostic, not a SHAP or multi-seed attribution, and we present it only as an indication of which fields that particular expert relies upon.

Two observations survive that caveat. First, kinematic features retain the dominant share of gain even inside this ERA5-inclusive expert: storm direction and speed are the two highest-ranked features at both horizons, which underscores that atmospheric information supplements rather than displaces kinematic signal. Second, every ERA5 field’s share of gain grows markedly from 24 to 48 h, with sea surface temperature rising from 1.66% to 4.15% and 200 hPa zonal wind from 1.34% to 3.32%, which

Table 7: Gain-based importance share of ERA5 fields within the environ-subset expert (held-out seed 0 diagnostic). Shares are fractions of total gain inside that expert only; storm-direction and storm-speed kinematic features rank above every ERA5 field and are omitted here.

ERA5 field	Share (24 h / 48 h)
Sea surface temperature (<i>sst</i>)	0.0166 / 0.0415
200 hPa zonal wind (<i>u200</i>)	0.0134 / 0.0332
Mean sea level pressure (<i>msl</i>)	0.0128 / 0.0323
Steering wind, zonal (<i>steer_u</i>)	0.0101 / 0.0161
500 hPa zonal wind (<i>u500</i>)	0.0086 / 0.0141
500 hPa meridional wind (<i>v500</i>)	0.0085 / 0.0137
Shear, zonal (<i>shear_u</i>)	0.0078 / 0.0152
850 hPa zonal wind (<i>u850</i>)	0.0070 / 0.0084
Shear, meridional (<i>shear_v</i>)	0.0067 / 0.0064
SST imputation flag (<i>sst_missing</i>)	0.0067 / 0.0140
850 hPa meridional wind (<i>v850</i>)	0.0048 / 0.0074
200 hPa meridional wind (<i>v200</i>)	0.0044 / 0.0156
Steering wind, meridional (<i>steer_v</i>)	0.0036 / 0.0135
Shear magnitude (<i>shear_mag</i>)	0.0032 / 0.0046

is an independent signature of the same lead-time dependence found in Section 6.3.

We deliberately do not interpret rank order within the shear and steering group. Those components are mutually collinear by construction, since shear is defined as a difference of the same wind levels that enter the steering term, so gain-based importance may apportion credit arbitrarily among them. Dropping a wind level or forcing a ridge penalty would remove the physical shear-steering decomposition the environ expert is designed to use; grouped permutation or SHAP analysis would be needed for stable within-block attribution. The substantive claim is the lead-time growth of total ERA5 gain (Section 6.3), not rank order within correlated winds.

Trajectory Validation — LAILA (2010): Held-out seed 0 vs IBTrACS

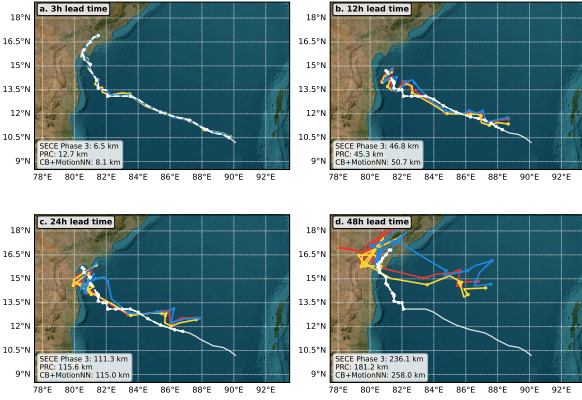


Figure 4: Cyclone Laila (2010) on held-out seed 0 (05_case_study_trajectories.csv). Four panels show 3, 12, 24, and 48 h lead times. White solid: storm centre at each origin; white dashed with markers: verified IBTrACS position at that lead. Coloured tracks are SECE Phase 3, PRC, and CB+MotionNN direct endpoints from each origin (not a rolled-out trajectory). Median great-circle errors in km appear in each panel.

6.5 Case studies

The examples below complement the held-out tables by showing how pooled rankings translate to individual storm geometry over the Bay of Bengal.

Fig. 4 shows predicted and observed tracks for a held-out storm. Each marker is a prediction issued from a distinct origin time along the trajectory and offset forward by the panel’s lead time, so the panels show direct multi-horizon endpoints rather than a recursively rolled-out track. For visual comparison we overlay SECE Phase 3, PRC, and CB+MotionNN on every panel; these are not necessarily the three lowest *pooled* medians at each horizon (Random Forest leads at 3 h, stacking at 12 and 24 h, and XGBoost at 48 h in Table 5), but they span the hybrid and ensemble families that remain competitive at long lead. The distinction between pooled leaderboard and single-storm behaviour matters in practice, as the following two cases demonstrate.

Cyclone Laila, 2010. Among the 88 test storms on the diagnostic seed, Laila exhibits the largest single-storm ERA5 benefit for the LightGBM diagnostic of Section 6.3: a 25.0 km improvement at 24 h and 107.0 km at 48 h when ERA5 features are included. We report this as an illustration of that mechanism, a plain gradient-boosted model’s response to atmospheric information, and not as a claim about SECE’s performance on this storm. Separately, at the ensemble level, SECE’s own per-storm error on Laila is 111.3 km at 24 h, locally better than the pooled-best competitor at 128.9 km,

while at 48 h the pattern reverses and XGBoost is locally stronger, 204.9 against 236.1 km, consistent with the pooled 48 h leaderboard.

A representative 1996 system. To illustrate typical rather than extreme behaviour we select the test storm whose SECE per-storm median error is closest to that seed’s overall SECE 24 h median of 106.43 km. This is an unnamed 1996 storm tracking through the central Bay of Bengal with genesis near 8.4°N, 87.7°E, whose per-storm median error of 107.24 km closely matches the typical-case benchmark. On this storm SECE at 107.2 km and the pooled-best competitor at 110.7 km are closely matched at 24 h, while XGBoost shows a wider local advantage at 48 h, 276.0 against 306.3 km. Both cases are consistent with the aggregate finding that SECE’s advantage concentrates at shorter horizons.

7 Discussion

The central empirical finding of this study is horizon-dependent. SECE’s context-aware, ensemble-of-experts architecture provides a statistically robust advantage at short to medium lead times, 3 and 12 h, where storm-level context such as recurvature proxies, coastal proximity, and atmospheric steering carries strong learnable signal. That advantage does not persist as statistically distinguishable superiority at 24 and 48 h, where SECE instead achieves parity with, and never significant inferiority to, the strongest available baselines.

We interpret this pattern from a bias-variance perspective on ensemble complexity. As target uncertainty grows with lead time, the marginal value of additional ensemble sophistication diminishes, and a comparatively simpler, well-regularised stacking approach becomes competitive. This extends the finding of Grinsztajn et al. [14], that tree-based methods outperform deep learning on tabular data at this scale, by one level: *within* the tree-based ensemble family, complexity itself should be calibrated to the information content available at a given forecast horizon rather than applied uniformly [13]. The effect sizes in Table 6 trace that calibration quantitatively, collapsing from -0.208 against stacking at 3 h to -0.071 at 24 h.

ERA5 features provide genuine value, but the locus of that value differs from where SECE’s architectural advantage is clearest. The per-storm LightGBM diagnostic of Section 6.3 places ERA5’s majority-of-storms benefit at 48 h, and the gain shares of Section 6.4 agree, roughly tripling from 24 to 48 h. The environ-expert architectural addition, by contrast, delivered its clearest ensemble-level gain at 24 h. We read this as genuine if imperfectly aligned confirmation that atmospheric information and ensemble architecture contribute complementary rather than identical improvements, and we flag the alignment between them as a direction

for further investigation rather than a discrepancy to be smoothed over.

For operational interpretation, the horizons at which SECE is significantly better are also the horizons that govern short-fuse decisions such as port closure and shelter activation, whereas 24 to 48 h guidance is better treated as a consensus problem in which several strong systems are statistically interchangeable [19, 12]. A practitioner choosing a single system for this basin could reasonably run SECE for short-range guidance and a simpler stack or persistence-residual model at long range, and the pooled-versus-seed narrative in Section 6.2 states that split explicitly rather than as a blanket recommendation.

Operational context clarifies what those medians do and do not imply. RSMC New Delhi verifies official tropical-cyclone track forecasts over the North Indian Ocean using WMO-standard direct position error against IMD best track [17]. For verified forecasts in 2009–2013, published mean errors were 124.1 km at 24 h and 202.1 km at 48 h [17]. Our held-out SECE pooled medians are 101.2 and 246.3 km at the same leads. Those figures are *not* directly comparable: we score retrospective IBTrACS origins with a fixed ML hindcast, whereas IMD errors average operational issuances from deep depression onward on a year-specific storm sample. We therefore do not claim to beat the operational multi-model ensemble or dynamical-statistical consensus products; this confirms benchmarks learned systems against each other. The comparison shows only that 24 h ML errors sit in the same order of magnitude as published official 24 h verification, while 48 h ML medians reflect the familiar difficulty of extended-range guidance in this basin.

Reading pooled, seed-level, and per-storm evidence together. Table 5 answers how systems order when all held-out test points are combined; the seed-level counts in Section 6.2 add whether that ordering is driven by one split or replicated across seeds, and whether paired tests support calling SECE better, worse, or tied at each horizon. Neither table, by itself, describes individual recurvature or landfall cases: the LightGBM ERA5 diagnostic (Section 6.3) and the Laila trajectories (Section 6.5) illustrate that a storm can be an ERA5 “win” under a single-model diagnostic while the full ensemble ordering on the same storm differs. We treat those layers as complementary rather than contradictory—pooled confirm for registry-style benchmarking, seed counts for stability, case panels for mechanism.

Placement among ML track studies. Most published North Indian Ocean track learners still compare few architectures on a single lead time or a single split [11, 20, 22]. Broader benchmarking ef-

forts emphasise protocol and sample design [7]; here we add multi-horizon, multi-seed confirm with Bonferroni testing, controlled ERA5 ablation, and an explicit negative result on spatial patches. Deep sequence baselines were scoped out after exploratory development under the same fixed training budget as the trees (Section 5.5), consistent with tabular-data evidence that gradient boosting remains strong at basin sample sizes [14]. Point ERA5 at the storm centre is a deliberate middle ground between kinematics-only models and gridded deep encoders [10]: lighter than full convolutional pipelines, yet sufficient for the per-storm and gain-share signals in Sections 6.3 and 6.4.

The leakage correction of Section 4 is equally part of that research contribution. Our pre-correction architecture appeared to win at every horizon on one split, and that sweep did not replicate once fresh seeds were drawn. Agreement between development and held-out tiers after freezing—246.94 versus 246.99 km at 48 h—is evidence that only a firewalled confirm can supply [36]. Iterative fusion-path choices driven by test-set rank tables can inflate ensemble benchmarks even when no learner sees a test label; we report that failure mode transparently so later work can separate architectural claims from evaluation hygiene.

8 Limitations and Future Work

Generalisation beyond the Bay of Bengal. Every model here is trained and confirmed on North Indian Ocean storms with Bay of Bengal landfall exposure. Feature subsets and router menus reflect Bay of Bengal synoptic steering (monsoon trough interaction, land proximity, funnel geometry). We report no zero-shot or retrained evaluation on the Western North Pacific, Atlantic, or other basins; cross-basin transfer—for example zero-shot scoring with NIO-trained models on typhoon tracks—remains strictly future work.

Observational quality across eras. Pre-1979 records show substantially reduced wind-observation coverage. Although the fixed-test control of Section 3.5 indicates that the wide window’s advantage on modern storms is not merely a test-composition artefact, we cannot fully exclude that some fraction of the observed gain reflects differences in how older, sparser-observation storms were originally tracked and recorded, as distinct from differences in the atmospheric information ERA5 provides for them.

Feature extent. ERA5 features here are extracted at a single point centred on the storm, or over a narrow spatial patch that we tested and did not adopt. Ocean subsurface heat content, higher vertical resolution, and genuine forward-looking operational NWP output as distinct from retrospective reanalysis were not incorporated and represent

natural extensions.

Point estimates. The systems evaluated here emit deterministic positions, which cannot express the growing uncertainty that dominates at long lead, precisely where our results show parity rather than separation. A probabilistic formulation trained under a proper scoring rule would be a better match to the long-range problem [41, 10].

Architectural directions not pursued. We identify, without implementing here, three directions with plausible mechanistic justification for narrowing the 24 to 48 h gap: physics-constrained trajectory modelling, in which displacement derives from a small set of learned physical parameters through a differentiable motion equation rather than being predicted directly; explicit predictive uncertainty modelling as noted above; and an end-to-end model attending jointly over storm history and spatial ERA5 fields rather than hand-engineered spatial statistics [23, 35, 10]. We regard these as second-stage directions building on the leakage-corrected, ERA5-validated foundation established here.

Single-seed attribution. The feature-importance analysis of Section 6.4 reflects one seed and one gain-based tree metric. A multi-seed, permutation- or SHAP-based analysis would strengthen mechanistic claims about which atmospheric fields drive the improvements, and is left to future work.

9 Conclusion

We present a leakage-corrected, statistically validated, ERA5-augmented benchmark for multi-horizon tropical cyclone track prediction over the Bay of Bengal. The proposed ensemble, SECE v2 Phase 3, significantly outperforms seven baseline systems at 3 and 12 h lead times and achieves statistical parity with the strongest baselines at 24 and 48 h, never falling to significant inferiority at any tested horizon among the evaluated tree-based systems. We identify and correct a test-set leakage failure mode encountered during our own development, document a controlled ablation isolating the contribution of ERA5 atmospheric features, and report negative results, specifically the rejection of spatial-patch features, alongside the positive ones. The intent is an honest account of what this class of model can and cannot currently achieve for an operationally important and underserved region.

For practitioners and follow-on research, the actionable reading is horizon-specific: routed ensembles merit priority at 3–12 h, where this confirm shows stable SECE leadership across seeds, whereas 24–48 h guidance is better treated as consensus among strong tree stacks until probabilistic or gridded ERA5 systems are tested under the same leakage-aware protocol. Tree ensembles at this scale

remain trainable on modest CPU hardware once re-analysis is pre-allocated, which keeps the approach relevant where full NWP capacity is limited. Probabilistic tracks, gridded encoders, and cross-basin transfer are natural extensions; they should adopt the same storm-wise, multi-seed discipline rather than repeating single-split leaderboard claims.

Acknowledgments

[Funding sources; Copernicus Climate Data Store for ERA5 access; computing support.]

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